addendum state the same rule, the two agree; where only this addendum states a rule, this addendum is the authority.

A1. COST SCALING ORDER

A unit cost is built in four ordered steps and each step rounds up to a whole tick before the next step reads it.

step 1	platform scaling	c1 = ceil(c0 * num / den)
step 2	regeneration	c2 = ceil(c1 * rnum / rden)
step 3	cold cache	c3 = ceil(c2 * knum / kden)
step 4	kind minimum	c4 = max(c3, minimum_cost_ticks)

c0 is base_cost_ticks from the component register. The step 1 fraction comes from the platform_scaling sheet of the capacity workbook, the step 2 and step 3 fractions from its cost_rules sheet, and the step 4 floor from build_cost_rules.jsonl.

Step 2 applies when the unit kind takes the regeneration surcharge and the unit carries contract_changed or generated_interface_rebuilt.

Step 3 applies when the unit kind takes the cold cache multiplier and the unit carries absent_from_previous_release.

Step 4 never lifts a unit whose base_cost_ticks is 0. A zero cost unit stays at cost 0 and is still a build unit.

Collapsing the steps into one multiplication with a single final rounding is not the documented arithmetic and does not give the documented result.

A2. CEILING DIVISION

For integers a >= 0 and b > 0, ceil(a / b) = (a + b - 1) // b.

No stage uses floating point. A fraction with denominator 0 is a row defect: the row is ignored and E_NON_INTEGER is recorded.

with the highest revision_seq. A budget whose scale_unit reads
milli_tick is converted with budget_ticks = ceil(budget_value / 1000).
The net budget of a pool is its budget_ticks less the reservation of
every active maintenance window naming that pool, each window converted
by its own field name, and never below 0.

A4. ADMISSION ORDER AND THE SELECTION OBJECTIVE
Pools are admitted in bytewise order of pool_id.
Inside one pool the candidate order is the tuple
(priority_class, closure_cost, unit_id)
ascending, compared bytewise on unit_id. closure_cost is the summed cost
of the candidate’s own prerequisite closure, counting only members not
already selected.
A candidate is admitted only with its whole prerequisite closure: the
candidate plus every dirty unit of the same target reachable from it
through required build edges. A closure that does not fit the remaining
net budget defers the candidate under E_CAPACITY and leaves the closure
members free to be admitted later on their own order position.
A closure exactly equal to the remaining net budget fits. A closure one
tick larger does not.
Ranking units one at a time by priority, without pricing the closure,
is not the documented objective.

A5. CROSS POOL PREREQUISITES
A prerequisite in another pool is judged after every pool has been
admitted. A selected unit whose cross pool required prerequisite is not
selected is demoted to deferred under E_PREREQ_DEFERRED, and the demotion
repeats until no selected unit has an unselected prerequisite. Capacity
released by a demotion is not readmitted to the pool.

A6. CYCLE BREAKS
A live cycle whose members form a breakable group loses one edge: the
edge minimising (break_cost, source, destination), compared bytewise on
the two identifiers. An edge with no declared break cost is not
removable. A live cycle in a blocking group blocks every target the
cycle is live on, and blocked units leave the schedule and the
deployment manifest under E_CYCLE_BLOCKING.

A7. SCHEDULE TIE BREAK
Among units ready at the same moment the order is (priority_class,
unit_id) ascending, with unit_id compared bytewise.

it names. A component named here takes the priority_class of its active row with the highest revision_seq, and the component register value applies to every component this table does not name.
A row whose state reads superseded is not applied. A row naming a component that is absent from the register is ignored under E_UNKNOWN_UNIT. A row whose component_id is not in canonical upper case form is ignored under E_ID_FORM.

component_id priority_class state revision_seq

C-0005	3	active	2
C-0007	3	active	2
C-0009	2	active	2
C-0011	2	active	2
C-0013	2	active	2
C-0015	1	active	2
C-0017	1	active	2
C-0019	1	active	2
C-0021	5	active	2
C-0023	5	active	2
C-0025	5	active	2
C-0027	4	active	2
C-0029	4	active	2
C-0031	4	active	2
C-0033	3	active	2
C-0035	3	active	2
C-0037	3	active	2
C-0039	2	active	2
C-0041	2	active	2
C-0043	2	active	2
C-0045	1	active	2
C-0047	1	active	2
C-0049	1	active	2
C-0051	5	active	2
C-0053	5	active	2
C-0055	5	active	2
C-0057	4	active	2
C-0059	4	active	2
C-0061	4	active	2
C-0063	3	active	2
C-0065	3	active	2
C-0067	3	active	2
C-0069	2	active	2
C-0071	2	active	2
C-0073	2	active	2
C-0075	1	active	2
C-0077	1	active	2
C-0079	1	active	2
C-0081	5	active	2
C-0083	5	active	2
C-0085	5	active	2
C-0087	4	active	2
C-0089	4	active	2
C-0091	4	active	2
C-0093	3	active	2
C-0095	3	active	2
C-0097	3	active	2
C-0099	2	active	2
C-0101	2	active	2
C-0103	2	active	2
C-0105	1	active	2
C-0107	1	active	2

C-0111	5	active	2
C-0113	5	active	2
C-0115	5	active	2
C-0117	4	active	2
C-0119	4	active	2
C-0121	4	active	2
C-0123	3	active	2
C-0125	3	active	2
C-0127	3	active	2
C-0129	2	active	2
C-0131	2	active	2
C-0133	2	active	2
C-0135	1	active	2
C-0137	1	active	2
C-0139	1	active	2
C-0141	5	active	2
C-0143	5	active	2
C-0145	5	active	2
C-0147	4	active	2
C-0149	4	active	2
C-0151	4	active	2
C-0153	3	active	2
C-0155	3	active	2
C-0157	3	active	2
C-0159	2	active	2
C-0161	2	active	2
C-0163	2	active	2
C-0165	1	active	2
C-0167	1	active	2
C-0169	1	active	2
C-0171	5	active	2
C-0173	5	active	2
C-0175	5	active	2
C-0177	4	active	2
C-0179	4	active	2
C-0181	4	active	2
C-0183	3	active	2
C-0185	3	active	2
C-0187	3	active	2
C-0189	2	active	2
C-0191	2	active	2
C-0193	2	active	2
C-0195	1	active	2
C-0197	1	active	2
C-0199	1	active	2
C-0201	5	active	2
C-0203	5	active	2
C-0205	5	active	2
C-0207	4	active	2
C-0209	4	active	2
C-0211	4	active	2
C-0213	3	active	2
C-0215	3	active	2

C-0219	2	active	2
C-0221	2	active	2
C-0223	2	active	2
C-0225	1	active	2
C-0227	1	active	2
C-0229	1	active	2
C-0231	5	active	2
C-0233	5	active	2
C-0235	5	active	2
C-0237	4	active	2
C-0239	4	active	2
C-0001	5	superseded	1
C-0015	5	superseded	1
C-0029	5	superseded	1
C-0043	5	superseded	1
C-4000	1	active	3
C-4001	1	active	3
C-4002	1	active	3
c-0003	1	active	4
c-0025	1	active	4
c-0047	1	active	4